Two Dimensions of Financial Trust: How Evaluative and Delegative Trust Shape Algorithm Aversion in Robo-Advisory

Chris [Last Name]*

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Abstract

[TODO: Write abstract (250 words max). Summarize the evaluative/delegative trust distinction, the dual-method design (CFPB NLP + Prolific experiment), and key findings.]

Keywords: trust, algorithm aversion, robo-advisors, delegative trust, evaluative trust, financial advice, behavioral finance, NLP, survey experiment
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*Corresponding author. [Affiliation]. Email: [email]. This paper was prepared for the UCLA Anderson Human × AI Finance Competition (Fink Center Conference on Financial Markets, April 2026). The author acknowledges extensive use of AI tools in research and writing; a detailed AI workflow description accompanies this submission.

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1 Introduction

[TODO: Write introduction. Motivate the research question, preview the evaluative/delegative trust framework, summarize the dual-method empirical strategy, and outline the paper's contributions.]

2 Theoretical Framework: Evaluative and Delegative Trust

A foundational finding of modern financial economics is that trust matters for economic outcomes. Guiso et al. (2004, 2008) demonstrate that generalized trust predicts stock market participation, portfolio allocation, and the use of formal financial instruments, even after controlling for risk aversion, wealth, and education. The mechanisms linking trust to financial behavior have since received sustained empirical attention: Sapienza and Zingales (2012) distinguish between generalized social trust and specific beliefs about institutional trustworthiness; Gennaioli et al. (2015) model financial advisors as "money doctors" whose primary service is providing trust that reduces clients' perceived variance of returns; and Gurun et al. (2018) demonstrate that geographic and cultural proximity — channels plausibly related to trust — predict advisory relationships and their outcomes.

Despite the richness of these empirical contributions, the literature shares a common structural assumption: trust is treated as unidimensional. It varies in *degree* — investors trust more or less — but not in *kind*. The standard measurement approaches reflect this assumption. The World Values Survey item used by Guiso et al. asks whether "most people can be trusted" on a binary scale. The trust game paradigm introduced by Berg et al. (1995) measures trust as the dollar amount sent to an anonymous counterparty — a continuous quantity representing the magnitude of risk accepted. Mayer et al.'s (1995) influential organizational framework decomposes trust into ability, benevolence, and integrity, but treats each as a continuous attribute of the trustee, not as a qualitatively distinct mode of the trust relationship itself.

These measurement approaches share a deeper assumption that Hardin (1993) identifies as a feature of "street-level" trust epistemology: initial trust is a learned Bayesian capacity that the truster brings *to* the encounter, relatively independent of the specific trustee's properties. Before we accumulate evidence about a particular advisor or institution, we rely on priors learned from a category of interaction — prior experiences with financial professionals, narratives about algorithmic decision-making, cultural attitudes toward institutional authority. Hardin's insight reframes the measurement problem. Instruments that present a novel counterparty and measure the truster's response are not

capturing a reaction to that counterparty's characteristics; they are measuring the truster's prior disposition toward the category. The question is not merely whether these instruments capture enough *variance* in trust — it is whether they capture the right *kind*. A truster who approaches algorithms as tools to be assessed and a truster who approaches them as agents to be entrusted with discretion are not expressing different magnitudes of the same attitude. They are operating in different modes, learned through different histories of encounter, and the standard instruments register both as a single quantity.

This unidimensional treatment creates an explanatory gap. Consider two well-documented patterns in algorithmic finance. First, Dietvorst et al. (2015) find that individuals who observe an algorithm err — even when it outperforms human judgment on the same task — subsequently prefer human forecasters. This "algorithm aversion" appears to reflect a fragile form of trust in which a single negative signal triggers exit. Second, and apparently in tension, Logg et al. (2019) find that individuals systematically *prefer* algorithmic to human advice in predictive domains, exhibiting what they term "algorithm appreciation." A unidimensional model treats these as contradictory findings requiring reconciliation through moderator variables: perhaps aversion appears in some contexts and appreciation in others, depending on task type, outcome observability, or individual differences.

We propose that the contradiction is an artifact of dimensional collapse. Algorithm appreciation and algorithm aversion are not opposing ends of a single trust continuum; they are manifestations of different trust *modes* operating under different conditions. Investors who evaluate algorithmic performance against benchmarks are engaged in a qualitatively different trust relationship from investors who grant an algorithmic advisor discretionary authority over their portfolio. The former — which we term *evaluative trust* — is inherently conditional on continued satisfactory performance and produces rapid exit when performance falters. The latter — which we term *delegative trust* — involves a transfer of judgment that is not reducible to performance monitoring and produces a qualitatively different set of responses to adverse events, including both greater resilience under some conditions and a distinctive *betrayal* reaction under others.

The distinction between evaluative and delegative trust is not merely taxonomic. It generates predictions that the unidimensional model cannot. If trust varies only in degree, then higher-trust investors should uniformly exhibit greater resilience after adverse performance: they trusted more, so they tolerate more. But practitioners and emerging empirical evidence suggest a more complex pattern. Some investors with strong pre-shock trust relationships are among the *most* reactive to performance failures, displaying emotional responses — a sense of violated expectations, of misplaced faith — that exceed what a rational updating model would predict. The unidimensional framework has no

vocabulary for this phenomenon. The evaluative/delegative framework does: these investors extended *delegative* trust, and its violation produces not the disappointment of a failed prediction but the betrayal of a breached relationship.

2.1 The Distinction

The evaluative/delegative framework draws on a substantial body of philosophical work on the nature of trust, applying it to financial relationships for the first time.

2.1.1 Evaluative Trust

Evaluative trust is the mode of trust that operates through assessment, verification, and conditional continuation. An investor operating in evaluative trust mode monitors portfolio performance, compares returns against relevant benchmarks, examines the rationale for specific allocation decisions, and maintains the advisory relationship conditional on continued evidence of competent management. The relationship is governed by evidence: when evidence supports the trustee's competence, the relationship continues; when evidence no longer supports it, the investor withdraws.

This mode of trust maps to well-established constructs in both philosophy and organizational behavior. Hardin (2002) models trust as "encapsulated interest": A trusts B because A believes that B's interests encapsulate A's interests — B will perform well for A because doing so serves B's own goals. The trust is grounded in a prediction about behavior derived from an assessment of incentive alignment, and it is withdrawn when the incentive structure no longer supports the prediction. McAllister (1995) identifies a cognition-based dimension of interpersonal trust in organizations, grounded in evidence of the trustee's competence and reliability, which he distinguishes from an affect-based dimension grounded in emotional bonds and mutual care. Evaluative trust in financial contexts aligns closely with McAllister's cognition-based construct: it is rational, evidence-conditional, and does not require emotional attachment to the trustee.

Evaluative trust is not inherently shallow or deficient. Sophisticated evaluative trust — in which an investor develops nuanced frameworks for assessing risk-adjusted performance, understands factor exposures, and monitors for style drift — represents a high level of financial engagement. A client who retains an advisor because they have carefully evaluated the advisor's strategy and believe it sound is exercising evaluative trust. The key feature is not the depth of engagement but the *conditionality*: the relationship is maintained because and insofar as the evidence supports it.

The behavioral signatures of evaluative trust follow directly from its structure. Because evaluative trust is conditional on evidence, its withdrawal is proportional to the strength of disconfirming evidence and takes the form of *disappointment* — a cognitive assessment that expectations were not met. The investor's response to a performance shock is to seek information (Was the underperformance systematic or idiosyncratic? How did comparable strategies perform?), reassess the relationship, and either revise expectations or exit. The emotional register is analytical: the investor feels let down, not betrayed.

2.1.2 Delegative Trust

Delegative trust involves a qualitatively different relationship. An investor operating in delegative trust mode has granted the advisor — whether human or algorithmic — discretionary authority over decisions that the investor cannot or chooses not to fully evaluate. The investor has, in a meaningful sense, *transferred judgment*: not merely delegated execution of a strategy the investor has approved, but ceded the authority to determine what strategy is appropriate, to adjust it as conditions change, and to exercise professional judgment about matters the investor does not independently assess.

This mode of trust draws on several philosophical accounts that have not previously been applied to financial relationships. Baier (1986) argues that trust inherently involves accepted vulnerability: to trust is to be in a position where the trusted party can harm you, and to accept that position rather than eliminate it through control or monitoring. Nickel (2012), analyzing trust in medical contexts, characterizes it as entrusting someone with discretionary authority — authority to make decisions using their own judgment, not merely to execute pre-specified instructions. Hawley (2014) adds that trust involves a *commitment*: to trust someone is to rely on their commitment to do what they are relied upon to do, and to feel betrayal when that commitment is violated. The trusted party is not merely predicted to perform; they are counted on to fulfill an obligation.

Holton (1994) provides the critical behavioral distinction. He argues that what differentiates trust from mere reliance is the *reactive attitude* of the trustor upon violation. When I rely on my alarm clock and it fails, I am annoyed — but I do not feel betrayed, because the clock made no commitment and I invested no vulnerability. When I trust my financial advisor and she makes unauthorized trades, I feel betrayed — because I granted her discretionary authority, accepted the vulnerability that comes with not monitoring every decision, and she violated the commitment that this acceptance implied. The presence or absence of betrayal as a reactive attitude is, for Holton, the diagnostic marker of whether a relationship constitutes trust or mere reliance.

Applied to financial relationships, delegative trust manifests as acceptance of opacity

(the investor does not insist on understanding every decision), tolerance of short-term uncertainty (the investor does not react to every quarterly fluctuation), and a distinctive *scope* of the relationship that extends beyond narrow performance metrics. The finding by Rossi and Utkus (2020) that advisory clients cite "peace of mind" and "delegation" as primary reasons for hiring advisors points toward delegative trust, though the existing literature does not isolate it from evaluative trust.

The behavioral signatures of delegative trust differ sharply from those of evaluative trust, particularly in response to adverse events. Because the investor has accepted vulnerability and granted discretionary authority, violation produces not proportional disappointment but *betrayal* — a response whose intensity reflects the depth of the trust relationship rather than the magnitude of the performance shortfall. The betrayal response is qualitatively distinct: the investor feels not merely that their predictions were wrong, but that their vulnerability was exploited, that the commitment they relied upon was hollow. This is the phenomenon that the unidimensional model cannot capture — the investor who was *most* trusting in the delegative sense may produce the *most* intense negative reaction, because they had the most to lose.

A further distinction concerns what happens *between* adverse events. An evaluative trustor monitors continuously; their trust is maintained through ongoing verification. A delegative trustor monitors less — not from inattention or ignorance, but from a *choice* to refrain from verification that would convert the relationship from delegation to surveillance. This distinction generates a testable prediction: platforms that encourage frequent performance monitoring may inadvertently shift users from delegative to evaluative trust mode, increasing the likelihood of exit after subsequent performance shocks. The investor who checks their portfolio daily is, through the act of checking, operating in evaluative mode — regardless of how the relationship was initially framed.

Delegative trust also has a distinctive failure mode that is not simply the inverse of its formation. Hardin (1993) observes that learned extreme distrust is self-reinforcing: it forecloses the encounters that might update the prior. A person who has developed generalized distrust of a category of agent avoids interactions with members of that category, ensuring they never accumulate the experience that could revise the disposition. Applied to our framework, this produces a phenomenon we term *delegative foreclosure*: the closure of an entire trust mode toward a category of trustee, not the downward revision of a trust estimate within that mode. A consumer who has had a negative experience with one algorithmic advisor — or who has absorbed sufficiently negative narratives about the category — does not merely lower their evaluative assessment of algorithmic advice. They foreclose the *possibility* of extending delegative trust to any algorithmic system. The

evaluative channel remains open: they can still be shown performance data, benchmark comparisons, and track records, and they may revise their evaluative trust accordingly. But the delegative channel is closed. No amount of performance evidence can reopen it, because performance evidence addresses the wrong dimension.

Delegative foreclosure is categorically different from algorithm aversion as the existing literature describes it. Dietvorst-style aversion is an evaluative phenomenon: the user observed poor performance and updated their competence assessment downward. Foreclosure is a structural change in the truster's available modes of relating to algorithmic trustees. This distinction matters practically because the interventions appropriate to each are different. Evaluative aversion can be addressed through better performance data, transparency, and track records. Foreclosure requires a different kind of repair — the reconstruction of conditions under which the truster can experience vulnerability as safe, which is a relational project, not an informational one.

2.1.3 Formal Characterization

The preceding accounts can be stated with greater precision. Let investor i's trust relationship with advisor j in financial domain D be characterized by an ordered pair (E, δ), where each component captures a distinct dimension of the relationship.

Definition 1 (Evaluative Trust). $E(i, j, D) \in [0, 1]$ is investor i's evidence-conditional assessment of advisor j's competence in domain D. Let s_t denote a performance signal received at time t (e.g., portfolio returns relative to a benchmark). Evaluative trust updates in response to performance information:

$$E_t = f(E_{t-1}, s_t), \quad \text{where } \frac{\partial f}{\partial s} > 0.$$

The function f is monotonically increasing in the signal: positive performance signals raise evaluative trust; negative signals lower it. When E_t falls below investor i's continuation threshold E^*, the investor exits. The reactive attitude upon exit is *disappointment*, proportional to the magnitude of the shortfall $(E^* - E_t)$.

Definition 2 (Delegative Trust). $\delta(i, j, D) \in [0, 1]$ is the degree to which investor i has granted advisor j discretionary authority over decisions in D that i does not independently verify. Unlike evaluative trust, delegative trust is not updated by performance signals alone:

$$\frac{\partial \delta}{\partial s_t} \approx 0 \quad \text{for ordinary performance signals.}$$

Delegative trust responds instead to *relational signals* — evidence about the commitment

structure itself. Let r_t denote a relational signal (e.g., discovery of a fiduciary breach, unauthorized exercise of discretion, institutional restructuring that alters the authority relationship). Then:

$$\delta_t = h(\delta_{t-1}, r_t), \quad \text{where } \frac{\partial h}{\partial r} \text{ is large and negative for commitment violations.}$$

The reactive attitude upon violation is *betrayal*, whose intensity reflects the prior level of delegative trust: a high-δ investor who experiences a relational violation produces a stronger negative response than a high-E investor who experiences an equivalent performance shock, because what has been violated is not a prediction but a grant of vulnerability.

These definitions generate three claims that the subsequent empirical analyses test.

Claim 1 (Dimensional Separability). E and δ respond to different signal classes. Across contexts that vary the salience of performance information versus relational commitment, E and δ should vary independently. Formally: a manipulation that increases the salience of performance monitoring (e.g., benchmark framing) should raise E without proportionally raising δ, while a manipulation that emphasizes fiduciary commitment should raise δ without proportionally raising E. A unidimensional construct would require both to move together.

Claim 2 (Asymmetric Post-Shock Response). After an adverse event $s_t < \bar{s}$:

> High-E investors: $\Pr(\text{exit}) = g(\bar{s} - s_t, E_{t-1})$ — exit probability increasing in shortfall magnitude.
>
> High-δ investors: $\Pr(\text{exit})$ is low when s_t carries only performance information, because $\partial\delta/\partial s_t \approx 0$. But if s_t also conveys relational information r_t (the advisor failed to exercise genuine judgment, the institutional commitment was hollow), then $\Pr(\text{exit})$ can *exceed* the evaluative case, accompanied by betrayal rather than disappointment.

This generates the betrayal premium prediction: for investors in the delegative mode, the gap between betrayal intensity and disappointment intensity should be positive after an adverse event, while for evaluative investors it should be near zero or negative.

Claim 3 (The Transparency Paradox). Let τ represent the degree of transparency or auditability in the advisory relationship. Increasing τ shifts the trust relationship along

different dimensions:

$$\frac{\partial E}{\partial \tau} > 0 \quad \text{(transparency makes performance verifiable, supporting evaluative trust)}$$
$$\frac{\partial \delta}{\partial \tau} < 0 \quad \text{(transparency eliminates the discretionary space in which delegative trust operates)}$$

This is the mechanism underlying O'Neill's (2002) diagnosis that audit culture undermines trust, formalized through the two-dimensional framework. A unidimensional model predicts that transparency uniformly increases trust; the two-dimensional model predicts that it increases one component while decreasing the other — with net effects on investor behavior that depend on which mode dominates the relationship.

2.1.4 The Relationship Between the Two Modes

The formal characterization makes explicit that evaluative and delegative trust are distinct dimensions, not opposite poles of a single continuum. In principle, an investor can exhibit high levels of both: carefully evaluating an advisor's track record (high E) while simultaneously granting discretionary authority over tactical decisions within an approved strategic framework (high δ). An investor can also exhibit low levels of both — maintaining a financial relationship through inertia without either evaluating performance or granting meaningful discretionary authority.

However, the two modes are not fully independent in practice. In constrained communicative contexts — including the complaint narratives analyzed in this study — the two modes compete for expression. A narrative dominated by relational language (betrayal, violated expectations, exploited vulnerability) leaves less space for analytical language (benchmark comparison, performance metrics, cost assessment), and vice versa. This produces an inverse correlation in observed expression that reflects the communicative constraint rather than a structural opposition between the constructs themselves. The critical test of dimensional separability is not the within-context correlation but the *cross-context asymmetry* predicted by Claim 1: whether the two dimensions vary independently across different conditions. As we demonstrate in the empirical analysis, delegative trust expression varies substantially across financial product categories while evaluative trust expression remains essentially constant — a pattern that a single bipolar dimension cannot produce.

The distinction between evaluative and delegative trust also clarifies a long-standing measurement problem. McAllister's (1995) affect-based trust items — developed for interpersonal workplace relationships — fail when applied to algorithmic trustees because

they operationalize the relational dimension of trust through interpersonal warmth ("We have a sharing relationship"; "I can talk freely to this individual about difficulties"). This failure has been interpreted as evidence that algorithms cannot receive trust beyond the competence-based dimension. We argue that the conclusion is premature. The failure is one of *operationalization*, not of *construct*: delegative trust with algorithmic systems operates not through affective bonds but through institutional features — fiduciary framing, regulatory protections, discretionary authority structures — that create the conditions for vulnerability acceptance and judgment transfer without requiring interpersonal warmth. In terms of the formal characterization, institutional features can raise δ even when affective channels are absent, provided they create credible commitment structures. The Prolific experiment reported in this paper tests this claim directly.

2.2 The Agency Theory Bridge

The evaluative/delegative distinction maps onto a foundational tension in financial economics: the spectrum between complete and incomplete contracting. Under Jensen and Meckling's (1976) framework, the principal–agent relationship is governed by a contract that specifies observable performance metrics, monitoring mechanisms, and enforcement provisions. The principal's problem is one of verification — ensuring the agent acts as contracted — and the solution is better information. This is evaluative trust rendered in the language of corporate finance: the investor trusts the advisor because and insofar as performance is observable, contractible, and enforceable.

But real advisory relationships are not complete contracts. Grossman and Hart (1986) demonstrate that when contingencies cannot be fully specified in advance — when the space of possible market states exceeds the contracting parties' ability to enumerate them — residual control rights must be allocated. Someone must decide what to do in circumstances the contract does not address. Delegative trust, in this framing, is the investor's acceptance of residual control rights held by the advisor: the authority to exercise judgment over decisions that were not and could not have been contracted upon. The investor does not merely delegate execution of a pre-specified strategy; they cede the right to determine what strategy is appropriate as conditions change. The trust inheres not in the ability to verify outcomes but in the willingness to grant discretion over uncontracted states of the world.

This mapping carries a methodological implication. The trust game introduced by Berg et al. (1995) — the dominant experimental paradigm for measuring trust in economics — is structurally a complete-contracting environment. The payoff matrix is known,

the interaction is one-shot, the counterparty is anonymous, and the decision reduces to a single monetary transfer with a calculable expected return. These are precisely the conditions under which evaluative trust operates: the trustor assesses the probability of reciprocation, calibrates the amount sent accordingly, and withdraws from the interaction once the outcome is observed. There is no residual control, no unspecified contingency, no discretionary authority to grant. If the finance literature validates its trust measures against trust game behavior — and it does extensively (Johnson and Mislin, 2011) — then the empirical foundation may be systematically measuring evaluative trust while the advisory relationships it seeks to explain are substantially delegative.

2.3 The "Peace of Mind" Reinterpretation

The evaluative/delegative framework's explanatory power is most clearly demonstrated against the most influential model of trust in financial advice. Gennaioli et al. (2015) propose that financial advisors function as "money doctors" whose primary service is the provision of trust. In their model, trust reduces the client's *perceived* variance of returns: the advisor does not necessarily improve expected outcomes, but the client experiences the portfolio as less risky because a trusted professional is managing it. Clients pay advisory fees not for alpha but for "peace of mind." The model is elegant, empirically productive, and — we argue — ambiguous on a dimension that determines its behavioral predictions.

The ambiguity is this: what kind of trust reduces perceived variance? Under an evaluative interpretation, the advisor provides peace of mind through optimization and legibility. The client monitors less frequently not because they have transferred judgment but because the advisor's demonstrated competence has lowered the expected incidence of events worth monitoring. The client's comfort is evidence-conditional: it persists because performance has been satisfactory and would dissolve if performance deteriorated. Under a delegative interpretation, peace of mind arises from the act of judgment transfer itself. The client experiences reduced variance not because outcomes are objectively smoother but because the burden of decision-making — of choosing, second-guessing, and bearing responsibility for allocation choices — has been lifted. The comfort comes from the relationship, not from verification of the relationship's outputs.

These interpretations generate sharply different predictions about what happens after a performance shock. The evaluative peace-of-mind client experiences the shock as a variance event that exceeds prior expectations. Their response is to reassess: Was the advisor's strategy flawed? Did the underperformance reflect systematic failure or mar-

ket conditions? The emotional register is disappointment calibrated to the magnitude of the shortfall relative to expectations. If the evidence warrants it, they exit; if it does not, they revise expectations and continue. The delegative peace-of-mind client experiences the shock differently. If the advisory relationship remains intact — if the advisor communicates, explains the approach going forward, and continues to exercise the discretionary authority the client granted — the client's response may be patience. The advisor is navigating difficulty on the client's behalf, which is precisely what they were trusted to do. But if the shock reveals that the delegative relationship was hollow — that the advisor was not exercising genuine judgment, or that the institutional structure does not support the discretionary authority the client believed they had granted — the response is betrayal, not disappointment. The intensity of the reaction reflects the depth of the prior vulnerability, not the magnitude of the financial loss.

Survey evidence suggests the delegative interpretation better describes advisory clients' self-reported experience. Rossi and Utkus (2020) find that clients cite "delegation" and "peace of mind" as primary reasons for retaining advisors — language that maps more naturally to judgment transfer than to variance assessment. But the existing literature does not isolate the two mechanisms. The Money Doctors model is agnostic about the *kind* of trust that does the variance-reducing work, and its predictions are correspondingly silent about the qualitative differences in post-shock behavior that the evaluative/delegative distinction reveals. Our framework does not replace the Gennaioli et al. model; it disaggregates a parameter the model treats as unitary.

2.4 Reconciling Algorithm Appreciation and Aversion

Three findings in the algorithmic decision-making literature appear to be in tension. Logg et al. (2019) find that individuals systematically prefer algorithmic to human advice in predictive domains, exhibiting "algorithm appreciation." Dietvorst et al. (2015) find that individuals who observe an algorithm err subsequently refuse to use it, even when it outperforms human forecasters — "algorithm aversion." And Bigman and Gray (2018) find that individuals resist algorithmic involvement in decisions with moral dimensions, even when the algorithm's judgments are equivalent to or better than human ones. The literature has treated these as context-dependent findings requiring reconciliation through moderator variables: appreciation prevails in some task environments, aversion in others, depending on outcome observability, task subjectivity, and the moral valence of the decision.

The evaluative/delegative framework offers a more parsimonious resolution. Algo-

rithm appreciation is evaluative trust operating as expected: when an algorithm demonstrates predictive competence in a domain where performance is observable and quantifiable, evaluative trust increases. The algorithm is assessed, verified, and preferred precisely because it outperforms. Algorithm aversion after errors is evaluative trust failing as expected: the same performance-conditional relationship that produced appreciation now produces withdrawal when a negative signal arrives. The trust was never resilient to error because it was never grounded in anything beyond continued evidence of competence. Both findings describe evaluative trust under different performance conditions, not opposing orientations toward algorithms.

Resistance to algorithmic moral decisions is a different phenomenon entirely. Moral decisions require precisely the exercise of discretionary judgment that defines delegative trust: they cannot be reduced to optimization over a known objective function, they require weighing incommensurable values, and they implicate the decision-maker's character — their integrity, not merely their competence. Bigman and Gray's finding reflects resistance to extending *delegative* trust to algorithms, not a generalized aversion. Respondents who willingly delegate predictive tasks to an algorithm (evaluative trust) balk at delegating moral judgment to the same system (delegative trust). The trust mode, not the trustee, has changed.

The concept of delegative foreclosure introduced above explains a further puzzle in this literature: the persistence of algorithm aversion even when users are presented with compelling evidence of algorithmic superiority. If the resistance were purely evaluative — a low competence assessment — then strong performance data should resolve it. Dietvorst et al. (2015) find that it does not. Our framework identifies the mechanism: for users who have foreclosed delegative trust toward algorithms, performance evidence addresses the evaluative dimension while the operative barrier is structural — the delegative channel is closed, and no volume of evidence within the evaluative channel can reopen it. The appropriate intervention is not more data but the reconstruction of conditions under which delegation feels safe.

Dietvorst et al.'s (2018) follow-up study provides further support. They find that even trivial modification ability — the capacity to adjust an algorithm's output by a small amount before it takes effect — substantially reduces algorithm aversion. This result is puzzling under a pure competence account: a minor adjustment does not meaningfully improve the algorithm's performance, so why should it change the user's willingness to rely on it? Under the evaluative/delegative framework, the answer is immediate. Modification grants the user a form of discretionary authority over the final output. In Nickel's (2012) terms, the user is no longer a passive recipient of algorithmic decisions but

a participant who exercises judgment — however minimally — over the outcome. This converts the relationship from pure evaluation (observing and assessing the algorithm's performance) to partial delegation (sharing authority over the final decision). The trust changes in kind, not merely in degree, and the behavioral consequences follow: a user who has exercised even token authority over a decision is less likely to experience the algorithm's error as a failure imposed upon them and more likely to experience it as a joint outcome for which they bear partial responsibility. **[VERIFY: Confirm that Dietvorst et al. (2018) is the correct citation for the modification-ability follow-up, and whether the effect was characterized as "trivial" or "small" modification.]**

2.5 The Transparency Paradox and Hypotheses

The evaluative/delegative framework generates a counterintuitive prediction about a central policy question in financial regulation: the role of transparency. O'Neill (2002, 2014) argues that the contemporary culture of transparency — the proliferation of disclosure mandates, performance metrics, audit requirements, and accountability mechanisms — has failed to restore public trust in institutions despite being designed expressly for that purpose. Her diagnosis is that transparency and audit culture cannot solve a trust crisis, and may in fact deepen it. The mechanisms of accountability, intended to make institutions more trustworthy, have instead produced "a culture of suspicion" in which institutional actors optimize for measurability rather than the underlying goods they are meant to deliver.

O'Neill's diagnosis is acute but theoretically incomplete. She identifies the failure without fully specifying the mechanism by which transparency undermines trust. The evaluative/delegative framework supplies this mechanism. Transparency builds evaluative trust: it makes the trustee's actions observable, verifiable, and assessable against explicit criteria. Disclosure mandates, benchmark reporting, and performance analytics all serve to make the advisory relationship more legible to the investor. But transparency simultaneously erodes the conditions under which delegative trust operates. Delegative trust requires discretionary space — the authority to exercise judgment over matters that the principal has chosen not to fully monitor. When every decision must be disclosed, justified, and subjected to audit, the discretionary space collapses. The advisor no longer holds residual control rights; they execute under surveillance. And an investor who verifies every decision is, by definition, operating in evaluative mode — regardless of how the relationship was initially framed or intended.

Applied to algorithmic finance, this analysis yields a specific prediction: explainability

mandates for robo-advisors — regulatory requirements that algorithms disclose the reasoning behind their investment decisions — may paradoxically *increase* algorithm aversion rather than reduce it. By making algorithmic decisions transparent and auditable, regulators shift users from a delegative trust equilibrium (in which they accept the algorithm's discretionary authority and tolerate short-term uncertainty) to an evaluative trust equilibrium (in which they assess each decision against their own expectations and benchmarks). The shift increases the user's sensitivity to short-term performance fluctuations and lowers the threshold for exit — precisely the opposite of the regulatory intent. Mandatory performance disclosure may induce monitoring behavior that converts the long-term commitment benefits documented by Gennaioli et al. (2015) into the short-term reactivity documented by Dietvorst et al. (2015).

Carter et al. (2020) identified the need for philosophical trust theory in financial contexts, applying Baier's vulnerability account and Hawley's commitment framework to the 2008 banking crisis. Their analysis demonstrated that reactive attitudes — betrayal, resentment, a sense of violated expectations — distinguish trust failures from mere reliance failures in financial relationships. But their contribution remained at the level of diagnosis: they showed that philosophical trust concepts apply to finance without operationalizing the distinction, without testing whether institutional framing shifts investors between trust modes, and without extending the framework to the algorithmic context that now dominates retail financial advice. The present paper answers all three.

2.5.1 Hypotheses

The theoretical framework developed across the preceding sections generates five preregistered predictions, tested through a between-subjects experiment in which participants are randomly assigned to one of three conditions — fiduciary framing (Condition A), performance framing (Condition B), or a neutral control (Condition C) — before encountering an identical adverse performance event.

H1 (Trust Mode Separation). Fiduciary framing produces higher delegative trust than performance framing; performance framing produces higher evaluative trust than fiduciary framing. If evaluative and delegative trust are distinct dimensions responsive to different institutional features, then framing that emphasizes discretionary authority and relational commitment (Condition A) should elevate delegative trust, while framing that emphasizes verifiable performance and benchmark comparison (Condition B) should elevate evaluative trust.

H2 (Betrayal Premium). The difference between betrayal and disappointment responses after the adverse event is larger under fiduciary framing than under performance

framing. Delegative trust violation produces betrayal (Holton, 1994); evaluative trust failure produces disappointment. If Condition A successfully induces delegative trust, the betrayal response should be elevated relative to disappointment — a pattern we term the "betrayal premium."

H3 (Algorithm Aversion Mechanism). The tendency to switch from algorithmic to human advice after the adverse event is predicted by evaluative trust scores but not by delegative trust scores. Algorithm aversion, in the Dietvorst sense, is an evaluative trust phenomenon: performance-conditional trust withdrawn after a negative signal. Delegative trust, grounded in accepted vulnerability and granted discretion rather than in performance verification, should not predict the algorithm-to-human switch.

H4 (Delegative Resilience). Higher delegative trust scores predict lower exit-seeking behavior after the adverse event, controlling for evaluative trust. Delegative trust tolerates short-term uncertainty because the trustor has accepted the trustee's exercise of discretionary authority through adverse conditions — the advisor is navigating difficulty, which is what they were trusted to do.

H5 (Financial Literacy Interaction). The treatment effect of fiduciary versus performance framing on delegative trust is concentrated among high-literacy respondents. Low-literacy respondents default to elevated delegative scores across conditions — not from trust, but from the absence of evaluative capacity. High-literacy respondents *can* evaluate but, under fiduciary framing, are given institutional permission not to. The interaction distinguishes genuine delegation — a choice made by those with the capacity to do otherwise — from delegation by default.

3 Data and Methods

[TODO: Write data and methods section. Describe CFPB complaint data (sample construction, NLP classification approach, dual-model validation) and Prolific experiment design (three conditions, measures, pre-registration).]

4 CFPB Complaint Analysis: Results

The CFPB analysis provides naturalistic evidence for dimensional separability. Where the Prolific experiment tests whether trust dimensions predict distinct post-shock responses in a controlled setting, the CFPB analysis asks whether evaluative and delegative trust are distinguishable in the language of real consumer complaints — filed by people with

actual financial grievances, not experimental participants responding to vignettes. The two analyses are methodologically independent: different populations, different product domains, different measurement approaches, and different analytic strategies. Neither depends on the other. This is the triangulation.

4.1 Sample and Classification Procedure

We sampled 9,000 consumer complaint narratives from the CFPB database, equally distributed across three product categories: mortgage ($n = 3{,}000$), checking and savings ($n = 3{,}000$), and credit card ($n = 3{,}000$). These categories were selected to vary in the degree to which the consumer–institution relationship involves delegation of discretionary authority. Mortgage relationships are long-duration, high-stakes, and involve ongoing servicing decisions that consumers cannot easily monitor — conditions that our theoretical framework identifies as conducive to delegative trust. Credit card relationships, by contrast, are largely transactional: the consumer evaluates terms, monitors statements, and assesses performance on observable metrics. Checking and savings accounts fall between these poles.

Each narrative was independently classified by two large language models — Claude Sonnet (Anthropic) and GPT-4o (OpenAI) — using identical prompts that defined the evaluative and delegative trust constructs and asked for (a) a categorical classification (evaluative, delegative, or unclassifiable), (b) continuous scores on each dimension (0–1), and (c) a confidence rating (0–1). Of the 9,000 narratives, Sonnet produced valid classifications for 8,991 (99.9%) and GPT-4o for 8,943 (99.4%). Dual-model analyses use the $N = 8{,}943$ narratives classified by both models.

4.2 Cross-Product Delegative Gradient

The central prediction is that delegative trust indicators should vary across product categories in a theoretically predictable way: highest for mortgages (long-term, opaque servicing), intermediate for checking and savings (ongoing relationship, moderate complexity), and lowest for credit cards (transactional, transparent terms). Evaluative trust indicators, by contrast, should be relatively uniform, since all three product types involve assessable performance dimensions.

The predicted monotonic ordering — mortgage $>$ checking/savings $>$ credit card — holds for delegative indicators across all three analysis variants: Sonnet-only, GPT-4o-only, and concordant cases (Table 1). The gradient is robust. For mortgage versus credit card, the delegative score difference is statistically significant and of moderate size:

Table 1: Cross-product delegative gradient: classification proportions and continuous scores by product category

	Mortgage ($n = 3{,}000$)	Checking/Savings ($n = 3{,}000$)	Credit Card ($n = 3{,}000$)
Panel A: Delegative classification proportion			
Sonnet	21.1%	16.4%	12.3%
GPT-4o	19.8%	13.2%	8.8%
Concordant only	16.7%	11.9%	7.1%
Panel B: Mean delegative score (SD)			
Sonnet	0.335 (0.249)	0.300 (0.228)	0.263 (0.210)
GPT-4o	0.411 (0.219)	0.371 (0.194)	0.329 (0.169)
Concordant avg.	0.347 (0.220)	0.310 (0.196)	0.269 (0.163)
Panel C: Mean evaluative score (SD)			
Sonnet	0.718 (0.210)	0.708 (0.220)	0.711 (0.233)
GPT-4o	0.724 (0.195)	0.705 (0.207)	0.731 (0.212)
Concordant avg.	0.749 (0.188)	0.736 (0.204)	0.749 (0.212)

Note. Each product category comprises $n = 3{,}000$ sampled narratives. Sonnet = Claude Sonnet (Anthropic); GPT-4o = GPT-4o (OpenAI). Concordant cases are narratives where both models assign the same categorical classification ($N = 7{,}369$). Delegative and evaluative scores range from 0 to 1. Concordant avg. reports the mean of Sonnet and GPT-4o continuous scores for concordant cases.

Sonnet $d = 0.313$ ($U = 5{,}413{,}026$, $p < .001$); GPT-4o $d = 0.419$ ($U = 5{,}545{,}242$, $p < .001$); concordant averaged scores $d = 0.399$ ($U = 3{,}882{,}984$, $p < .001$).

Evaluative scores, by contrast, are essentially flat across products. The mortgage–credit card difference in evaluative scores is negligible: Sonnet $d = 0.030$; GPT-4o $d = -0.031$; concordant $d = 0.001$ (n.s., $p = .145$). This asymmetry is precisely what dimensional separability predicts. If the two trust modes were a single underlying construct — "complaint intensity" or "relationship quality" — we would expect both dimensions to co-vary across product types. Instead, only delegative trust tracks the theoretically relevant product dimension (relationship opacity and duration), while evaluative trust remains stable.

4.3 Dual-Model Concordance

A classification scheme is only useful if it can be applied reliably.

Table 2: Confidence-stratified inter-model concordance ($N = 8{,}943$)

Confidence tercile	Range	N	Agreement	Cohen's κ
Low	0.51–0.78	3,759	68.8%	0.356
Medium	0.78–0.83	3,124	87.7%	0.652
High	0.85–0.97	2,060	99.2%	0.970
Overall	—	8,943	82.4%	0.556

Note. Confidence is the mean of Sonnet and GPT-4o confidence ratings for each narrative. Tercile boundaries are based on the distribution of mean confidence scores. Agreement = percentage of narratives receiving the same categorical classification (evaluative, delegative, or unclassifiable) from both models. Cohen's κ corrects for chance agreement.

Overall categorical agreement is 82.4% ($\kappa = 0.556$, moderate; Table 2). This headline figure, however, masks a sharp confidence–concordance gradient. When both models assign high confidence, agreement reaches 99.2% ($\kappa = 0.970$, almost perfect). When confidence is low, agreement drops to 68.8% ($\kappa = 0.356$). This pattern is theoretically informative: the narratives that are difficult for classifiers are the ambiguous cases where trust mode is genuinely unclear, not cases where the classification scheme is arbitrary.

Continuous scores show strong cross-model convergence. Sonnet and GPT-4o delegative scores correlate at $r = .703$ (Spearman $\rho = .738$, $p < .001$); evaluative scores at $r = .680$ ($\rho = .713$, $p < .001$). Both models independently recover the same latent structure in the complaint narratives.

4.4 Moral Foundations Validation

As an independent check on construct validity, we scored all narratives using the extended Moral Foundations Dictionary (eMFD) and compared moral language across trust classifications. The theoretical prediction is specific: delegative trust violations should activate relational moral foundations — particularly loyalty (the foundation most closely associated with trust betrayal) — more than evaluative trust violations.

Narratives classified as delegative scored significantly higher than evaluative narratives on all five moral foundations, with the largest effects for loyalty ($d = 0.267$, $p < .001$) and sanctity ($d = 0.263$, $p < .001$), followed by care ($d = 0.235$, $p < .001$). The loyalty result is particularly important: in the Moral Foundations framework, the loyalty foundation tracks in-group betrayal and allegiance — precisely the relational vocabulary our theory associates with delegative trust violation.

Table 3: eMFD correlations with trust scores: dual-model comparison (Spearman ρ)

	Delegative score		Evaluative score	
Foundation	Sonnet	GPT-4o	Sonnet	GPT-4o
Care	.099***	.129***	−.112***	−.120***
Fairness	.069***	.106***	−.086***	−.103***
Loyalty	.118***	.158***	−.108***	−.128***
Authority	.035**	.074***	−.062***	−.078***
Sanctity	.108***	.155***	−.119***	−.139***

Note. $N = 8{,}943$. Moral foundation scores derived from the extended Moral Foundations Dictionary (eMFD). Delegative and evaluative scores are continuous (0–1) as assigned by each model.
* $p < .05$, ** $p < .01$, *** $p < .001$.

Correlation patterns reinforce this interpretation (Table 3). Delegative scores correlate positively with all foundations, most strongly with loyalty ($\rho = .118$, $p < .001$). Evaluative scores correlate *negatively* with all foundations, most strongly with sanctity ($\rho = -.119$, $p < .001$) and care ($\rho = -.112$, $p < .001$). These opposing sign patterns hold for both Sonnet and GPT-4o classifications, providing further evidence that the two models are recovering the same construct distinction.

4.5 Human Validation

To assess whether the AI classifications capture distinctions recognizable to a human coder, one author (the first author) independently classified a confidence-stratified subsample of 100 narratives. Stratification ensured representation across confidence levels:

30 low-confidence, 40 medium, and 30 high-confidence narratives (based on Sonnet confidence bins).

Categorical agreement was fair by conventional benchmarks: $\kappa = 0.264$ for human–Sonnet and $\kappa = 0.361$ for human–GPT-4o. The two AI models agreed more with each other ($\kappa = 0.590$) than either did with the human coder. However, the confidence-stratified pattern is dramatic and informative. At high confidence, human–Sonnet agreement reaches 93.3% and human–GPT-4o reaches 90.0%. At low confidence, agreement drops to 30.0% and 50.0%, respectively.

The primary source of disagreement is systematic and interpretable: the human coder classified 56% of narratives as unclassifiable, compared to 21% (Sonnet) and 33% (GPT-4o). The most common disagreement pattern — 23 of 51 disagreement cases — was both AI models classifying a narrative as evaluative where the human coded it as unclassifiable. This reflects a higher threshold for inferring trust language in the human coding: factual complaints that describe service failures without explicit evaluative language (e.g., "they put a hold on my account and I can't access it") were coded unclassifiable by the human but evaluative by the models, which inferred frustration with performance from the complaint structure.

Continuous score correlations tell a more encouraging story. Human and model delegative scores correlate moderately (human–Sonnet $r = .488$, $p < .001$; human–GPT-4o $r = .511$, $p < .001$), and both are substantially lower than the inter-model correlation ($r = .776$). Evaluative score correlations follow the same hierarchy (human–Sonnet $r = .308$; human–GPT-4o $r = .402$; inter-model $r = .779$). The models agree with each other more than with the human, but the human–model agreement is significant and in the expected direction, particularly at high confidence.

We interpret this pattern as follows. The dual-model approach functions as an automated coding system with known biases (over-attribution of evaluative intent in factual narratives). The high-confidence subset, where all three raters converge, represents the clearest cases of each trust mode. The analyses reported above are robust to restricting to concordant cases (see Table 1, concordant rows), which partially addresses this concern.

4.6 Dimensional Relationship

Our theoretical framework characterizes evaluative and delegative trust as independent dimensions. In the Prolific experiment, the two composites correlate at $r = .246$ (6% shared variance; Table 6), consistent with dimensional separability. The CFPB data present a more complex picture.

Across the full sample, delegative and evaluative continuous scores are moderately negatively correlated (Sonnet $r = -.592$; GPT-4o $r = -.574$; concordant averaged $r = -.561$). This is substantially stronger than the Prolific correlation and in the opposite direction. Three features of the data suggest this reflects properties of the CFPB sample rather than the constructs themselves.

First, the correlation varies substantially across product types: mortgage $r = -.807$, checking/savings $r = -.589$, credit card $r = -.400$ (Sonnet). If the negative correlation reflected a true inverse relationship between the constructs, we would expect it to be stable across contexts. Instead, it tracks product complexity — strongest where the complaint space is most constrained (mortgages, where nearly every complaint involves either servicing failures or relationship violations) and weakest where complaints are more heterogeneous (credit cards).

Second, the scatter plot (Figure 1) reveals that the correlation is partly driven by clustering at classification-typical regions rather than a smooth negative slope. Narratives cluster in a high-evaluative/low-delegative region and a high-delegative/moderate-evaluative region, with relatively sparse representation in the joint-high quadrant. This is consistent with the classification mechanics: when a narrative strongly expresses one trust mode, the classifier assigns lower scores to the other mode, producing a mechanical negative correlation within the classification system.

Third, and most importantly, the CFPB sample is censored: it consists entirely of people who have filed formal complaints. These consumers have already experienced a breakdown in their relationship with the institution. In the language of our framework, the sample is heavily weighted toward cases where at least one trust mode has been violated. In a general population, individuals may hold high levels of both evaluative and delegative trust simultaneously (as the Prolific data suggest). In the CFPB sample, the complaint context compresses the joint distribution toward the violated region of the trust space.

We therefore treat the negative correlation as an informative feature of the complaint context — consistent with the theory's prediction that trust violations are mode-specific — rather than evidence against dimensional independence. The stronger test of independence comes from the Prolific general-population sample, where the weak positive correlation ($r = .246$, 6% shared variance) supports separability.

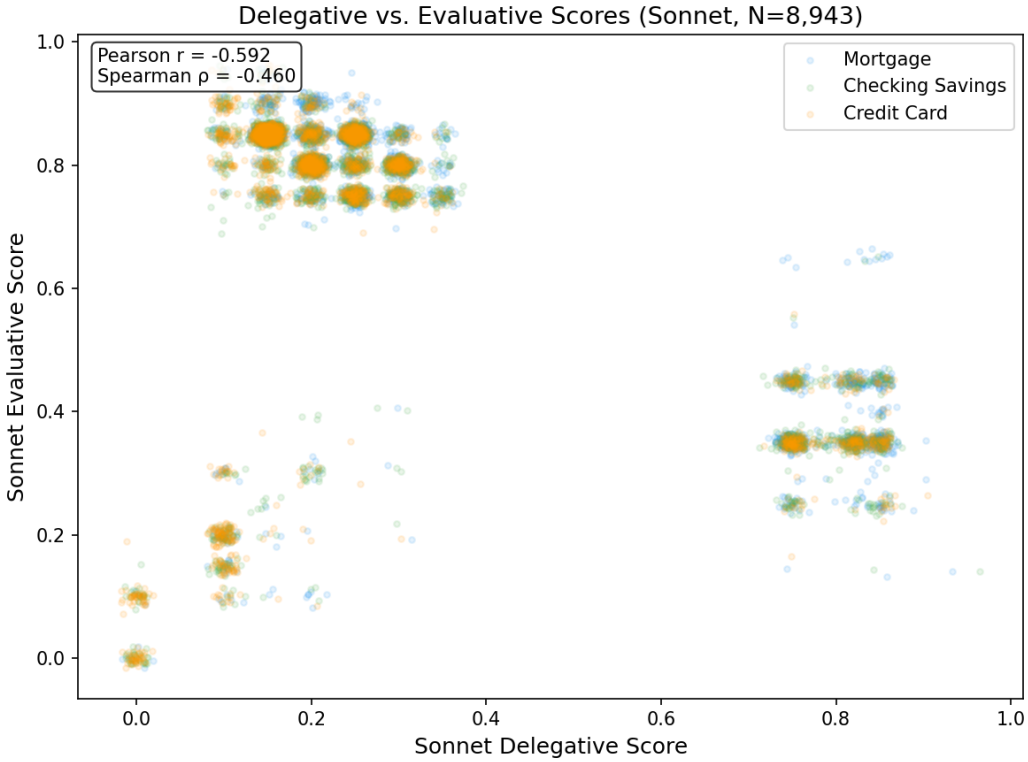

Figure 1: Delegative versus evaluative trust scores (Sonnet classifications, $N = 8{,}943$). Points are colored by product type. The moderate negative correlation ($r = -.592$) is partly driven by clustering at classification-typical regions rather than a smooth inverse relationship.

4.7 Robustness: Narrative Length

Delegative narratives are longer than evaluative narratives across all product types (e.g., mortgage: delegative $M = 2{,}144$ characters vs. evaluative $M = 1{,}590$). This raises a potential confound: perhaps longer narratives simply provide more text for classifiers to identify relational cues, and the delegative classification is partly an artifact of narrative length.

To address this, we regressed Sonnet delegative scores on log-transformed character count and product category dummies. Length is a significant predictor ($b = 0.079$, $p < .001$). However, the cross-product delegative gradient retains 74.3% of its magnitude after controlling for length (raw gradient: 0.071; residualized gradient: 0.053).[1] The theoretically meaningful between-product variation in delegative trust cannot be explained by narrative length alone.

4.8 Exploratory: Company-Level Variation

As an exploratory analysis, we examined whether specific mortgage companies generate disproportionately more delegative complaints. Among the 15 companies with 50 or more mortgage narratives, delegative classification proportions ranged from 3.6% to 37.8% — a spread of 34.2 percentage points. This variation suggests that institutional behavior may drive the trust mode activated in consumer complaints, a hypothesis consistent with our framework's claim that delegative trust responds to relational rather than performance signals. We return to this finding in Section 6.

4.9 Summary

The CFPB analysis provides naturalistic evidence for three aspects of our theoretical framework. First, the cross-product delegative gradient — consistent across two independent AI classifiers and robust to narrative length controls — demonstrates that trust modes are separable in real consumer behavior and track theoretically predicted product characteristics. Second, the eMFD analysis shows that delegative complaints activate relational moral vocabulary, particularly loyalty, at higher rates than evaluative complaints. Third, the dual-model approach, validated against human coding at high confidence levels, provides a reliable method for identifying trust modes in unstructured text at scale.

The CFPB database does not cover investment advisory services, which fall under SEC and FINRA jurisdiction rather than CFPB oversight. This means we cannot directly study

[1] Gradient values computed from unrounded mean delegative scores.

robo-advisor complaints. We treat this jurisdictional boundary as a triangulation feature rather than a limitation: the CFPB analysis covers banking products using naturalistic complaint data, while the Prolific experiment covers algorithmic investment advice using controlled vignettes. The theoretical framework's predictions hold across both domains, different populations, and different measurement approaches.

5 Prolific Experiment: Results

5.1 Sample and Exclusions

We recruited 471 responses through Prolific. Table 4 reports the pre-registered exclusion pipeline. After removing survey previews (3), pre-registered pilot responses (17), and matching to Prolific's approved submissions, 446 responses remained. We then applied two pre-registered exclusion criteria: completion time under 120 seconds removed 89 responses (19.9%), and the dual failure criterion — failing both the instructed attention check and the comprehension check — removed an additional 3 responses. The final analytical sample comprised 354 participants (Condition A: $n = 115$; Condition B: $n = 127$; Condition C: $n = 112$).

Table 4: Sample exclusion pipeline

Stage	N Remaining	N Excluded	Reason
Qualtrics total	471	—	Raw export
Remove previews	468	3	Survey preview responses
Remove pilots	451	17	Pre-registered pilot exclusion
Prolific matching	446	5	Unmatched to Prolific approved submissions
Speed exclusion	357	89	Completion time $<$ 120 seconds
Dual-check exclusion	354	3	Failed both attention and comprehension checks

The speed exclusion rate warrants comment. Approximately one in five participants completed the survey in under two minutes, a duration insufficient to read the vignette, process the performance notification, and respond thoughtfully to the dependent measures. Among the 357 participants who met the speed threshold, only 3 failed the dual attention-comprehension criterion, suggesting that the speed filter effectively separated engaged from disengaged respondents.

A condition balance check revealed a significant age imbalance: participants in Condition C were older ($M = 41.2$) than those in Condition A ($M = 37.0$) and Condition B ($M = 37.5$), $F(2, 351) = 4.15$, $p = .017$. All other covariates — financial literacy, gender,

education, income, and robo-advisor experience — were balanced across conditions. Age is included as a covariate in all regression models.

5.2 Descriptive Statistics

Table 5: Means and standard deviations of dependent variables by experimental condition

	Condition A ($n = 115$)	Condition B ($n = 127$)	Condition C ($n = 112$)	Overall ($N = 354$)
Evaluative trust ($\alpha = .43$)	5.43 (1.01)	5.28 (1.12)	5.19 (1.01)	5.30 (1.05)
Delegative trust ($\alpha = .81$)	3.84 (1.39)	4.04 (1.48)	3.86 (1.39)	3.92 (1.42)
Betrayal	4.23 (1.61)	4.02 (1.71)	4.24 (1.61)	4.16 (1.64)
Disappointment	5.84 (1.16)	5.71 (1.36)	5.71 (1.18)	5.75 (1.24)
Betrayal premium	−1.62 (1.57)	−1.69 (1.65)	−1.46 (1.63)	−1.59 (1.62)
General exit	4.68 (1.60)	4.57 (1.72)	4.76 (1.58)	4.67 (1.64)
Algorithm aversion	5.03 (1.53)	4.65 (1.58)	4.95 (1.43)	4.86 (1.52)

Note. All items measured on 7-point Likert scales (1 = Strongly Disagree, 7 = Strongly Agree). Betrayal premium = Betrayal minus Disappointment. Condition A = Fiduciary/Relationship framing; Condition B = Performance/Analytics framing; Condition C = Neutral control. Cronbach's α reported for composite measures.

Table 5 presents means and standard deviations for all dependent variables by condition. Two patterns merit attention. First, evaluative trust ($M = 5.30$, $SD = 1.05$) was substantially higher than delegative trust ($M = 3.92$, $SD = 1.42$) across all conditions. Participants were generally willing to monitor and evaluate the algorithm's performance but ambivalent about granting it discretionary authority — a pattern consistent with the framework's prediction that the default psychological posture toward algorithms is evaluative. Second, disappointment ($M = 5.75$) exceeded betrayal ($M = 4.16$) in every condition, producing a consistently negative betrayal premium ($M = -1.59$). After an adverse event, participants' dominant reactive attitude was disappointment at the outcome rather than betrayal by the agent, again consistent with evaluative rather than delegative trust governing the human–algorithm relationship.

The delegative trust composite showed good internal consistency (Cronbach's $\alpha = .807$). The evaluative trust composite, comprising only two items, showed poor internal consistency ($\alpha = .429$). We discuss implications of this measurement limitation below and in Section 6.

Table 6 presents bivariate correlations among the trust composites and dependent variables. The evaluative and delegative trust composites are positively correlated ($r =$

.246, $p < .001$), confirming that they are related constructs — as any two trust measures should be — while sharing only approximately 6% of variance. The two composites thus capture substantially distinct dimensions of the participant's orientation toward the algorithm. Their downstream profiles reinforce this conclusion. Delegative trust is negatively correlated with algorithm aversion ($r = -.313$, $p < .001$) and general exit ($r = -.155$, $p < .01$), but uncorrelated with betrayal ($r = .061$, n.s.). Evaluative trust shows a largely flat profile across dependent variables, consistent with both its lower reliability and the descriptive observation that evaluative engagement is near-uniformly high across the sample. Among the post-shock dependent variables, general exit and algorithm aversion are strongly correlated ($r = .690$, $p < .001$), reflecting their shared orientation toward abandoning the algorithm, though they differ in specificity: general exit captures platform-switching intent while algorithm aversion captures the preference for human advice specifically.

Table 6: Bivariate correlations among trust composites and dependent variables

	1	2	3	4	5	6	7
1. Evaluative trust	—						
2. Delegative trust	.246***	—					
3. Betrayal	.054	.061	—				
4. Disappointment	.113*	−.102	.399***	—			
5. Betrayal premium	−.032	.140**	.712***	−.359***	—		
6. General exit	−.015	−.155**	.578***	.406***	.278***	—	
7. Algorithm aversion	−.109*	−.313***	.515***	.420***	.202***	.690***	—

Note. $N = 354$. Evaluative trust = mean of Items 1–2; Delegative trust = mean of Items 3–5; Betrayal premium = Betrayal minus Disappointment.
* $p < .05$, ** $p < .01$, *** $p < .001$.

5.3 Experimental Manipulation

We predicted that framing the robo-advisor in fiduciary terms (Condition A) would produce higher delegative trust than framing it in performance terms (Condition B), and that the performance framing would produce higher evaluative trust (H1). Neither prediction was supported. One-way ANOVA revealed no significant differences across conditions in delegative trust, $F(2, 351) = 0.75$, $p = .473$, $\eta^2 = .004$, or evaluative trust, $F(2, 351) = 1.50$, $p = .225$, $\eta^2 = .009$ (Table 7). The observed means were directionally opposite to predictions on both measures, though far from significance.

The betrayal premium (H2) was likewise undifferentiated across conditions, $F(2, 351) =$

Table 7: One-way ANOVA results for experimental manipulation effects (H1, H2)

Hypothesis	DV	$F(2, 351)$	p	η^2	Predicted	Observed
H1a	Delegative trust	0.75	.473	.004	A > B	B > A (n.s.)
H1b	Evaluative trust	1.50	.225	.009	B > A	A > B (n.s.)
H2	Betrayal premium	0.57	.564	.003	A > B	A > B (n.s.)

Note. Condition A = Fiduciary/Relationship framing; Condition B = Performance/Analytics framing; Condition C = Neutral control. n.s. = not significant at $\alpha = .05$. $N = 354$.

0.57, $p = .564$, $\eta^2 = .003$. All three conditions produced negative betrayal premiums, indicating that disappointment dominated betrayal regardless of framing.

We interpret the manipulation failure in Section 6. Crucially, the null effect of condition assignment does not bear on the validity of the trust constructs as individual-difference measures. The primary predictions of the evaluative–delegative framework concern the downstream consequences of trust modes, not their experimental inducibility, and we test these predictions in the analyses that follow.

5.4 Delegative Trust and Post-Shock Resilience

Our framework predicts that delegative trust — granting the advisor discretionary authority with acceptance of opacity — should buffer against exit-seeking behavior after an adverse event, because the delegative truster's relationship with the advisor is not contingent on any particular performance outcome. H4 tests this prediction directly.

Table 8 reports the full regression model predicting general exit intent. The model is significant, $F(9, 344) = 2.18$, $p = .023$, $R^2 = .054$. As predicted, delegative trust negatively predicts exit intent ($B = -0.206$, $SE = 0.064$, $\beta = -0.179$, $p = .001$, 95% CI $[-0.33, -0.08]$). Each unit increase in delegative trust on the seven-point scale is associated with a 0.21-point decrease in the intention to research alternative platforms. Evaluative trust, by contrast, shows no predictive relationship with exit intent ($B = 0.037$, $p = .667$). The two trust dimensions thus make divergent predictions about post-shock behavior: one predicts resilience, the other is inert.

Two additional findings are worth noting. Comprehension check pass positively predicts exit intent ($B = 0.616$, $p = .008$): participants who correctly identified the performance shortfall were more likely to seek alternatives. This pattern is consistent with evaluative processing of the adverse event — the negative signal was encoded and acted upon — though the effect is captured by the check variable rather than the evaluative trust composite, a point to which we return below. Financial literacy approaches signif-

Table 8: OLS regression predicting general exit intent (H4)

Predictor	B	SE	β	t	p	95% CI
Delegative trust	−0.206**	**0.064**	−0.179	−3.23	**.001**	**[−0.33, −0.08]**
Evaluative trust	0.037	0.087	0.024	0.43	.667	[−0.13, 0.21]
Condition A (Fiduciary)	−0.056	0.218	−0.016	−0.26	.797	[−0.48, 0.37]
Condition B (Performance)	−0.132	0.211	−0.039	−0.62	.532	[−0.55, 0.28]
Age	0.005	0.007	0.040	0.74	.460	[−0.01, 0.02]
Financial literacy	−0.251	0.134	−0.106	−1.87	.062	[−0.51, 0.01]
Robo-advisor experience	0.242	0.266	0.049	0.91	.365	[−0.28, 0.76]
Comprehension pass	0.616**	0.232	0.146	2.65	.008	[0.16, 1.07]
Attention pass	0.577	0.478	0.064	1.21	.228	[−0.36, 1.52]
$R^2 = .054$, Adjusted $R^2 = .029$, $F(9, 344) = 2.18$, $p = .023$						

Note. Reference category for condition is C (Neutral control). Robo-advisor experience coded as 1 = has used a robo-advisor. Comprehension pass = correctly identified WealthPath underperformance. Attention pass = selected instructed response. $N = 354$.
* $p < .05$, ** $p < .01$, *** $p < .001$.

icance as a negative predictor ($B = -0.251$, $p = .062$), suggesting that higher-literacy participants may be more tolerant of short-term underperformance.

The result is robust to alternative exclusion criteria. Under strict attention exclusion ($N = 342$), the delegative trust coefficient is $B = -0.207$, $p = .001$. Under strict comprehension exclusion ($N = 289$), it is $B = -0.161$, $p = .021$. Direction and significance are unchanged.

5.5 Delegative Trust and Algorithm Aversion

H3 tested whether the trust dimensions predict algorithm aversion — the preference for a human advisor or self-management over the algorithm after the adverse event. We predicted that evaluative trust would drive algorithm aversion and that delegative trust would not. The results reversed this prediction in a theoretically informative way.

Table 9 reports the full model, which is highly significant, $F(9, 344) = 6.82$, $p < .001$, $R^2 = .151$, and represents the strongest overall fit of any regression in the experimental analyses. Delegative trust is the dominant predictor of algorithm aversion ($B = -0.326$, $SE = 0.056$, $\beta = -0.304$, $p < .001$, 95% CI $[-0.44, -0.22]$). Participants who scored higher on willingness to grant the algorithm discretionary authority were substantially less likely to prefer a human advisor after the adverse event. Evaluative trust, by contrast, was not significant ($B = -0.039$, $p = .614$).

This finding has a straightforward interpretation within the evaluative–delegative

Table 9: OLS regression predicting algorithm aversion (H3)

Predictor	B	SE	β	t	p	95% CI
Delegative trust	-0.326***	**0.056**	-0.304	-5.79	$< .001$	**[-0.44, -0.22]**
Evaluative trust	-0.039	0.077	-0.027	-0.50	.614	[-0.19, 0.11]
Condition A (Fiduciary)	0.073	0.192	0.023	0.38	.703	[-0.30, 0.45]
Condition B (Performance)	-0.250	0.186	-0.079	-1.35	.179	[-0.62, 0.12]
Age	-0.006	0.006	-0.051	-1.01	.315	[-0.02, 0.01]
Financial literacy	-0.191	0.118	-0.086	-1.62	.107	[-0.42, 0.04]
Robo-advisor experience	-0.215	0.234	-0.047	-0.92	.358	[-0.68, 0.25]
Comprehension pass	0.584**	0.204	0.149	2.86	.004	[0.18, 0.99]
Attention pass	1.179**	0.421	0.141	2.80	.005	[0.35, 2.01]
$R^2 = .151$, Adjusted $R^2 = .129$, $F(9, 344) = 6.82$, $p < .001$						

Note. Reference category for condition is C (Neutral control). Robo-advisor experience coded as 1 = has used a robo-advisor. Comprehension pass = correctly identified WealthPath underperformance. Attention pass = selected instructed response. $N = 354$.
* $p < .05$, ** $p < .01$, *** $p < .001$.

framework. The factor that protects against algorithm aversion is not a favorable performance assessment — it is the willingness to cede decisional authority. Participants who had granted the algorithm discretion, who had accepted opacity and declined to second-guess every decision, maintained that posture through an adverse event. Those who had not granted such discretion showed no corresponding resilience. This is dimensional separability operating in the predicted direction: the two trust modes engage different psychological mechanisms, and only the delegative mechanism predicts who stays with the algorithm after failure.

As in H4, the check variables are informative. Both comprehension pass ($B = 0.584$, $p = .004$) and attention pass ($B = 1.179$, $p = .005$) positively predict algorithm aversion. Participants who were paying attention and who correctly encoded the performance shortfall were more, not less, likely to want a human advisor. This pattern is consistent with evaluative processing — the participants who received and encoded the negative signal were the ones who responded by seeking alternatives — but the evaluative trust composite does not capture this effect. The poor internal consistency of the evaluative trust composite ($\alpha = .429$) likely explains the discrepancy: the comprehension and attention check variables are acting as proxies for evaluative engagement that the two-item scale measures inadequately. This is a limitation of the present study; we discuss it further in Section 6, but note here that the limitation biases against finding evaluative trust effects and does not threaten the delegative trust findings.

This result also held under both robustness checks. Strict attention exclusion ($N = 342$): delegative trust $B = -0.322$, evaluative trust $B = -0.058$ (n.s.). Strict comprehension exclusion ($N = 289$): delegative trust $B = -0.316$, evaluative trust $B = -0.047$ (n.s.).

5.6 Financial Literacy Interaction

H5 predicted that financial literacy would moderate the effect of condition on delegative trust, with high-literacy participants showing a larger difference between Conditions A and B. The interaction was not significant (Condition A $\times$ high literacy: $B = 0.072$, $p = .869$; Condition B $\times$ high literacy: $B = -0.109$, $p = .797$). This null follows from the manipulation failure: where condition assignment does not produce main effects on delegative trust, an interaction with literacy is unlikely to emerge.

One finding from the H5 model merits attention independent of the interaction. Prior robo-advisor experience was the strongest predictor of delegative trust in the model ($B = 0.729$, $p = .001$). Participants who had previously used a robo-advisor scored nearly three-quarters of a point higher on the delegative trust composite, regardless of experimental condition or financial literacy. This is a substantial effect that dwarfs any condition-level variation and is consistent with the philosophical account of trust developed in Section 2: trust postures are learned through repeated encounters rather than installed by informational framing. We return to this finding in Section 6.

5.7 Exploratory Analyses

Three pre-registered exploratory analyses are reported here. These were designated as exploratory in the pre-registration and should not be treated as confirmatory.

We predicted that Condition A would produce higher variance in delegative trust than the other conditions (E1), reflecting polarization between participants who can and cannot extend delegative trust to algorithms. This was not supported: Levene's test was not significant, $F(2, 351) = 0.32$, $p = .728$, and variances were similar across conditions (A: 1.94; B: 2.18; C: 1.92).

We predicted that comprehension check failures would show higher delegative trust and lower evaluative trust than passes (E2), on the theory that delegative trusters encode performance signals less carefully. The predicted directions emerged on both measures — failures showed slightly higher delegative trust ($M = 3.97$ vs. 3.90) and lower evaluative trust ($M = 5.11$ vs. 5.34) — but neither difference reached significance (delegative: $t =$

-0.33, $p = .739$; evaluative: $t = 1.60$, $p = .110$). Comprehension failure rates did not differ meaningfully by condition (A: 19.1%; B: 18.9%; C: 17.0%; Table 10).

Table 10: Comprehension check performance by experimental condition

Condition	Pass *N* (%)	Fail *N* (%)	Total
A (Fiduciary)	93 (80.9%)	22 (19.1%)	115
B (Performance)	103 (81.1%)	24 (18.9%)	127
C (Neutral)	93 (83.0%)	19 (17.0%)	112
Total	289 (81.6%)	65 (18.4%)	354

Note. Comprehension check asked participants to identify WealthPath's performance relative to the S&P 500. Correct answer: WealthPath performed worse than the market.

We tested whether prior robo-advisor experience moderated the relationship between condition and each dependent variable (E3). Two of fourteen interactions reached significance: Condition A $\times$ robo-experience on disappointment ($B = 1.174$, $p = .022$), and Condition B $\times$ robo-experience on the betrayal premium ($B = -1.313$, $p = .035$). Among participants with robo-advisor experience, fiduciary framing produced greater disappointment after the adverse event, while performance framing produced a more negative betrayal premium — that is, a stronger dominance of disappointment over betrayal. Both patterns are consistent with the framework: experienced users who are told the algorithm is their fiduciary may have activated higher expectations, while experienced users primed with performance framing process the shortfall as a pure evaluative failure. However, with fourteen interaction tests conducted at $\alpha = .05$, approximately 0.7 significant results are expected by chance. These two interactions modestly exceed chance expectation but would not survive correction for multiple comparisons. We treat them as suggestive and hypothesis-generating.

5.8 Summary

The experimental manipulation did not differentiate trust modes across conditions, but the evaluative–delegative distinction proves consequential when measured as a continuous individual difference. The two trust composites share only 6% of variance ($r = .246$) and show divergent downstream profiles (Table 6). Delegative trust is the primary predictor of resilience to both exit-seeking (H4: $\beta = -0.179$, $p = .001$) and algorithm aversion (H3: $\beta = -0.304$, $p < .001$) following an adverse event. Evaluative trust predicts neither outcome, though this null must be interpreted cautiously given the low reliability

of the two-item composite. The trust dimensions thus make divergent and specific predictions about post-shock behavior, supporting the core theoretical claim of dimensional separability. These findings complement the CFPB analysis reported in Section 4, which provides naturalistic evidence for the same constructs; we discuss the triangulation in Section 6.

6 Discussion

The results reported above provide convergent evidence from two independent methods that evaluative and delegative trust are separable dimensions with distinct behavioral signatures. They also reveal important boundary conditions on the framework — most notably, the failure to manipulate trust modes through vignette framing — that are themselves theoretically informative. We organize the discussion around four themes: what the manipulation failure teaches, the pattern of dimensional separability across methods, the reconciliation of the algorithm aversion literature, and the practical implications of delegative foreclosure.

6.1 What the Manipulation Failure Teaches

The most straightforward prediction of our framework — that fiduciary framing would elevate delegative trust and performance framing would elevate evaluative trust (H1) — was not supported. All between-condition comparisons were nonsignificant (all $p > .22$, all $\eta^2 < .01$). The manipulation did not shift participants between trust modes.

We do not treat this as evidence against the evaluative/delegative distinction. The distinction is supported by the individual-differences analyses (H3, H4) and by the CFPB naturalistic data, neither of which depends on the manipulation. But the failure requires explanation, and the explanation is substantive rather than merely methodological.

Hardin's (1993) street-level epistemology provides the key. If trust postures are learned Bayesian priors — dispositions the truster brings to the encounter based on accumulated experience with a category of interaction — then a brief vignette describing a robo-advisor in fiduciary terms cannot install a delegative posture in participants who have not learned, through experience, that algorithmic systems can be trusted with discretionary authority. The manipulation asks participants to *imagine* delegating judgment; it cannot provide the experiential grounding that delegative trust requires.

The robo-advisor experience predictor supports this interpretation directly. Prior experience with robo-advisory platforms was the strongest predictor of delegative trust in

the sample ($B = 0.729$, $p = .001$), substantially stronger than any experimental manipulation. Participants who had actually used algorithmic advisors — who had the experiential encounters Hardin's framework identifies as necessary for trust formation — exhibited higher delegative trust regardless of condition assignment. This is not a confound; it is the mechanism. Delegative trust toward algorithms forms through repeated encounters in which vulnerability is experienced as safe, not through institutional framing alone.

This finding has implications for both research and practice. Experimentally, it suggests that single-session vignette designs — the dominant paradigm in the algorithm aversion literature — may be structurally unable to study delegative trust formation, because they cannot provide the experiential history that delegative trust requires. Longitudinal designs, in which participants interact with algorithmic advisors over multiple sessions and encounter both favorable and adverse outcomes, may be necessary. For practitioners, the implication is that onboarding framing is insufficient: fiduciary language in terms of service or welcome screens is unlikely to build delegative trust without a sustained relationship that gives the framing experiential content.

6.2 Dimensional Separability Across Methods

The core theoretical claim — that evaluative and delegative trust are separable dimensions responding to different signal classes (Claim 1) — is supported by two independent lines of evidence.

In the CFPB analysis, the cross-product delegative gradient provides the cleanest test. Delegative trust indicators decline monotonically from mortgage (21.1%) to checking/savings (16.4%) to credit card (12.3%), tracking the theoretically predicted dimension of relationship opacity and duration. Evaluative trust indicators are flat across the same products (Sonnet $d = 0.030$ for mortgage vs. credit card). This asymmetric pattern — one dimension varying while the other holds constant — is the signature of dimensional separability. It is robust across two independent AI classifiers and survives control for narrative length (74.3% of the gradient retained).

In the Prolific experiment, separability appears in the predictive structure. Delegative trust predicts both exit intent (H4: $B = -0.206$, $p = .001$) and resistance to algorithm aversion (H3: $B = -0.326$, $p < .001$). Evaluative trust predicts neither (both $p > .6$). The two dimensions are not merely distinguishable in language — they predict different behaviors in a controlled post-shock context. The correlation between the two trust composites ($r = .246$, 6% shared variance) further supports separability: related but far from redundant.

The triangulation is strengthened by the methodological independence of these findings. The CFPB analysis uses naturalistic complaint text, AI-based classification, and a between-product design across banking products. The Prolific experiment uses self-report measures, a between-subjects vignette design, and an algorithmic investment context. The two analyses share no participants, no measurement instruments, no product domain, and no analytic strategy. That both converge on dimensional separability — through different operationalizations, in different populations, across different financial products — constitutes substantially stronger evidence than either alone.

6.3 Reconciling Algorithm Aversion and Appreciation

The evaluative/delegative framework resolves an apparent contradiction in the algorithmic decision-making literature that has persisted for over a decade.

Logg et al.'s (2019) algorithm appreciation and Dietvorst et al.'s (2015) algorithm aversion are not competing findings. Both document evaluative trust operating as our framework predicts: updating monotonically with performance signals. Logg et al.'s participants observed positive signals and increased trust; Dietvorst et al.'s participants observed negative signals and withdrew it. The disagreement in the literature arises from treating these as evidence about fundamentally different human–algorithm relationships, when they are movements along the same evaluative dimension under different evidentiary conditions.

Kormylo et al. (2026) introduce a third finding that completes the picture: betrayal aversion reduces delegation to human experts but not to algorithmic experts. Their explanation — that people do not experience betrayal toward machines — is empirically robust. Our framework provides the structural account of *why* the betrayal response is absent: betrayal is the reactive attitude specific to delegative trust violation, and delegative trust requires the attribution of discretionary authority to the trustee. Under default experimental conditions — including Kormylo et al.'s trust games and delegation tasks — participants approach algorithms as tools to be assessed, not agents to be entrusted with judgment. Delegative trust does not form, so the betrayal response has no ground to stand on.

The three findings thus decompose cleanly within the two-dimensional framework. Logg et al. (2019) document evaluative trust responding to positive signals. Dietvorst et al. (2015) document evaluative trust responding to negative signals. Kormylo et al. (2026) document the absence of delegative trust under default conditions. The literature has been studying variation *within* evaluative trust — appreciating or averting based on

observed performance — without recognizing that a second dimension exists.

Our Prolific data add to this picture. The H3 result — that delegative trust, not evaluative trust, is the dominant predictor of resistance to algorithm aversion — demonstrates that the second dimension is not merely a theoretical possibility but an active predictor of behavior. Participants with high delegative trust did not switch to human advice after the performance shock, even though they experienced the same adverse event as participants with low delegative trust. The evaluative dimension, which the existing literature treats as the complete explanation for algorithm aversion, had no predictive power ($p = .614$). This is precisely what the framework predicts: algorithm aversion is an evaluative phenomenon, but *resistance* to it comes from the delegative dimension. Investors who have granted discretionary authority do not revert to performance-based evaluation after a single adverse signal.

This reinterpretation also generates a novel prediction that none of the three papers can produce independently: if experimental designs create conditions for delegative trust formation toward algorithms — through fiduciary framing combined with sustained interaction — then betrayal-like responses should emerge even toward algorithmic advisors. Our manipulation attempted this and failed, likely because a single vignette cannot provide the experiential grounding required. The prediction remains testable in longitudinal designs.

6.4 Delegative Foreclosure and Design Implications

The concept of delegative foreclosure — introduced in the theoretical framework (Section 2) as the closure of an entire trust mode toward a category of trustee — connects the empirical findings to practical questions about robo-advisor design and financial regulation.

The persistence of algorithm aversion is a well-documented puzzle. Dietvorst et al. (2015) find that even showing users evidence of algorithmic superiority does not fully overcome aversion. The standard explanation appeals to evaluative mechanisms: perhaps users weight errors more heavily than successes, or perhaps they hold algorithms to higher standards than human advisors. Our framework offers a structural alternative. For users who have foreclosed delegative trust toward algorithms, the performance evidence is addressing the wrong dimension. Their evaluative channel may be open — they can acknowledge that the algorithm performs well on metrics — but the delegative channel is closed. They cannot bring themselves to grant the algorithm discretionary authority, to accept the vulnerability that delegation requires. No amount of performance

evidence can reopen that channel, because the closure is structural, not evidentiary.

This analysis inverts conventional wisdom about robo-advisor design. The industry's dominant strategy — increase transparency, provide performance dashboards, offer explainability tools — is optimized for evaluative trust. It makes the algorithm's decisions legible, verifiable, and assessable against benchmarks. Under our framework, this approach succeeds for users already in evaluative mode (who may appreciate the algorithm per Logg et al., 2019) but fails for users in delegative foreclosure (who need relational repair, not more data) and actively undermines nascent delegative trust in users who might otherwise develop it. This is the transparency paradox formalized in Claim 3: transparency builds evaluative trust ($\partial E/\partial \tau > 0$) but erodes the conditions under which delegative trust can form ($\partial \delta/\partial \tau < 0$).

The CFPB company-level analysis provides suggestive naturalistic evidence for this mechanism. Among mortgage companies with 50 or more complaints, the proportion of delegative complaints ranged from 3.6% to 37.8% — a 34.2 percentage-point spread. While exploratory, this variation suggests that institutional behavior — not just product type — shapes which trust mode consumers activate. Companies whose servicing practices create conditions for experienced vulnerability (long hold times on modification decisions, opaque escrow management, inconsistent communication) may generate more delegative complaints precisely because they have first created and then violated the conditions for delegative trust. This is a hypothesis for future research, not a conclusion from the current data.

The practical implication for robo-advisory platforms is specific: if the goal is long-term client retention rather than initial adoption, platforms should invest in fiduciary framing, discretionary authority structures, and the institutional conditions that permit delegative trust to form over time. This means less emphasis on real-time portfolio dashboards and per-trade explanations, and more emphasis on the relationship-like features that signal the platform's acceptance of fiduciary responsibility. The platforms that have best retained assets under management may already be doing this intuitively; our framework provides the theoretical basis for what they are doing and why it works.

6.5 Limitations

Several limitations qualify the conclusions drawn above.

First, the evaluative trust composite in the Prolific experiment demonstrated poor internal reliability ($\alpha = .429$, two items). This attenuates any effects of evaluative trust on dependent variables and means that the null findings for evaluative trust in the H3 and

H4 regressions cannot be interpreted as strong evidence of absence. The evaluative dimension may predict algorithm aversion and exit intent in ways our measurement was unable to detect. Future research should develop multi-item evaluative trust scales for algorithmic advisory contexts.

Second, the human validation of the CFPB classification involved a single coder (the first author) rating 100 narratives. While categorical agreement with the AI models was fair ($\kappa = 0.264$–0.361) and improved dramatically at high confidence (93.3% agreement), a single coder cannot establish inter-rater reliability in the conventional sense. The primary disagreement pattern — the human coding more narratives as unclassifiable — is systematic and interpretable, but future work should employ multiple independent human coders to better calibrate the AI classification system.

Third, the CFPB sample is censored toward consumers who have already experienced relationship breakdowns severe enough to motivate a formal complaint. The negative correlation between delegative and evaluative scores ($r = -.59$) likely reflects this censoring rather than a true inverse relationship between the constructs, as argued in the results section (Section 4). But the censored sample also means we observe trust modes primarily in their violated state, which limits what the data can tell us about trust formation and maintenance.

Fourth, the robo-advisor experience predictor — the strongest evidence for Hardin's experiential-learning mechanism — is correlational. We cannot rule out that pre-existing individual differences (risk tolerance, technology comfort, financial sophistication) drive both the decision to use robo-advisors and the tendency to report higher delegative trust, rather than the experience itself building delegative trust over time. Longitudinal designs that track trust formation through early interactions with algorithmic advisors would provide stronger evidence.

Fifth, the manipulation failure means that causal claims about the effect of institutional framing on trust mode are not supported by the experimental data. The individual-differences analyses (H3, H4) are associational: they show that delegative trust *predicts* behavioral outcomes, not that it *causes* them. The theoretical framework provides the causal architecture, and the CFPB naturalistic data provide ecological validity, but the experimental causal evidence that a fully pre-registered design was intended to provide is absent.

Finally, the CFPB database does not cover investment advisory services (SEC/FINRA jurisdiction), and the Prolific experiment uses hypothetical vignettes rather than real investment decisions with financial consequences. Neither study observes actual investor behavior in response to actual portfolio losses. The triangulation across methods partially

addresses this concern — naturalistic text and controlled experiment, banking products and algorithmic investment advice — but the ultimate test of the framework requires observation of real advisory relationships through adverse market conditions.

7 Conclusion

This paper introduced a two-dimensional framework for understanding trust in financial advisory relationships, distinguishing evaluative trust (evidence-conditional performance assessment) from delegative trust (the grant of discretionary authority with acceptance of opacity). Three empirical contributions support the framework's core claims.

First, dimensional separability is demonstrated across methods. In naturalistic CFPB complaint narratives, delegative trust indicators track product characteristics (relationship opacity and duration) while evaluative indicators remain flat — a pattern inconsistent with unidimensional models. In the Prolific experiment, the two trust composites share only 6% of their variance and predict non-overlapping sets of behavioral outcomes.

Second, delegative trust predicts resilience to both exit and algorithm aversion after adverse performance, while evaluative trust predicts neither. The H3 result — that delegative trust is the dominant predictor of resistance to algorithm aversion — reverses the pre-registered prediction but provides stronger evidence for the framework: algorithm aversion is an evaluative phenomenon, and resistance to it comes from the delegative dimension.

Third, the framework resolves a decade-long apparent contradiction in the algorithmic decision-making literature. Algorithm appreciation (Logg et al., 2019), algorithm aversion (Dietvorst et al., 2015), and the absence of betrayal aversion toward algorithms (Kormylo et al., 2026) are each manifestations of the evaluative dimension operating under different signal conditions. The second dimension — delegative trust — has been absent from the experimental literature because the standard paradigms do not create conditions for its formation.

The concept of delegative foreclosure, derived from the intersection of our framework with Hardin's (1993) street-level epistemology, identifies a mechanism the existing literature cannot name: the structural closure of a trust mode toward a category of trustee, distinct from the downward revision of a trust estimate. This concept explains both the persistence of algorithm aversion in the face of superior performance evidence and the failure of our own manipulation to shift trust modes through framing. Trust postures toward algorithmic advisors are learned through experience, not installed by institutional language. The robo-advisor experience predictor in the Prolific data ($B = 0.729$, $p = .001$)

provides correlational support for this experiential-learning mechanism.

For practitioners, the framework inverts a common design assumption. The robo-advisory industry's emphasis on transparency, explainability, and performance dashboards is optimized for evaluative trust — and may actively undermine the delegative trust that predicts long-term client retention. Platforms seeking to reduce algorithm aversion and exit should invest not in better performance communication but in the institutional conditions that permit delegative trust to form: fiduciary framing, discretionary authority structures, and sustained interaction that allows vulnerability to be experienced as safe.

For regulators, the transparency paradox formalized in Claim 3 poses a genuine dilemma. Explainability mandates that make algorithmic decisions more legible may shift users from delegative to evaluative trust equilibria, increasing sensitivity to short-term performance fluctuations and raising the probability of exit — the opposite of the regulatory intent. This does not argue against transparency as a value, but it suggests that disclosure regimes should be designed with attention to their effects on trust mode, not merely on information availability.

The paper's limitations — particularly the evaluative trust measurement problem and the manipulation failure — constrain the strength of causal claims. But the convergence of naturalistic and experimental evidence, across different populations, products, and measurement approaches, establishes the evaluative/delegative distinction as a productive framework for understanding trust in financial advisory relationships. The faithful machine is not the one that performs best on benchmarks. It is the one that earns, and honors, the grant of discretionary authority.

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A CFPB Classification Prompt

[TODO: Include the full CFPB classification prompt used for Sonnet and GPT-4o, with the evaluative/delegative trust coding rubric.]

B Prolific Survey Instrument

[TODO: Include the full Prolific survey instrument: vignettes for all three conditions, trust measure items, betrayal/disappointment scales, and demographic questions.]

C Inter-Rater Reliability Details

[TODO: Report inter-rater reliability details: Sonnet vs. GPT-4o agreement rates, Cohen's kappa, confusion matrices, and disagreement analysis.]

D Supplementary Tables

[TODO: Include supplementary tables: full product-level breakdowns, robustness checks, alternative specifications, and additional statistical tests.]